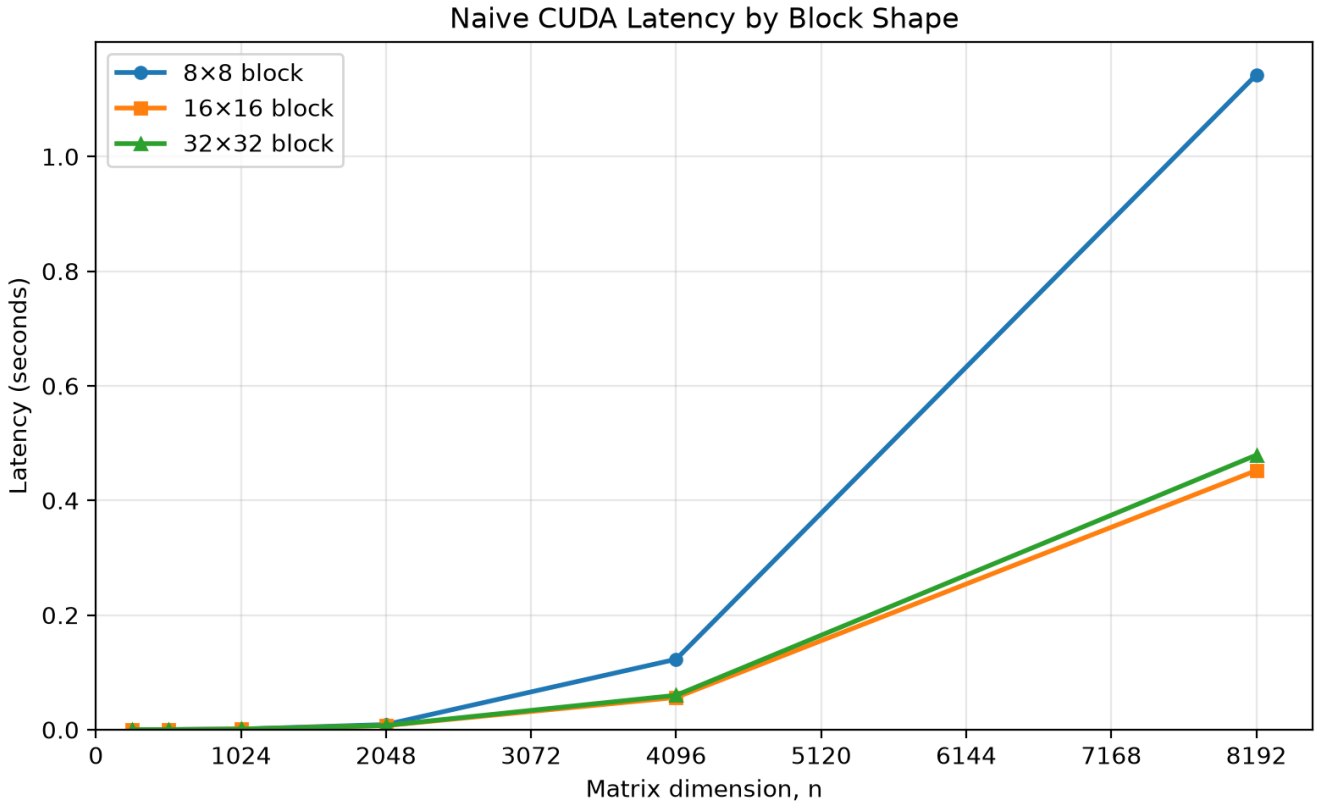
Section Overview

- Memory Boundedness, Basic Tiling, and Analysis

Let's do some experimenting

This kernel assigns one thread per output element, and computes them independently. So changing block size shouldn't matter right?



Blocks and SMs

Interpretation

This happens because entire **blocks** are scheduled to SM's. The larger block sizes of 16x16 and 32x32 lead to more **L1 cache hits**.

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Block shape	**L1 load hit rate**
8×8	83.51%
16×16	87.13%
32×32	94.98%

RTX 3090, $M = N = K = 4096$

Shared Memory (SMEM) Caching

Programmable L1 Cache (SMEM)

Remember how our launch config required us to group threads as blocks? That is because **blocks** are scheduled to SM's as a unit, so **threads in the same block have access to the SM's L1 cache/SMEM.**

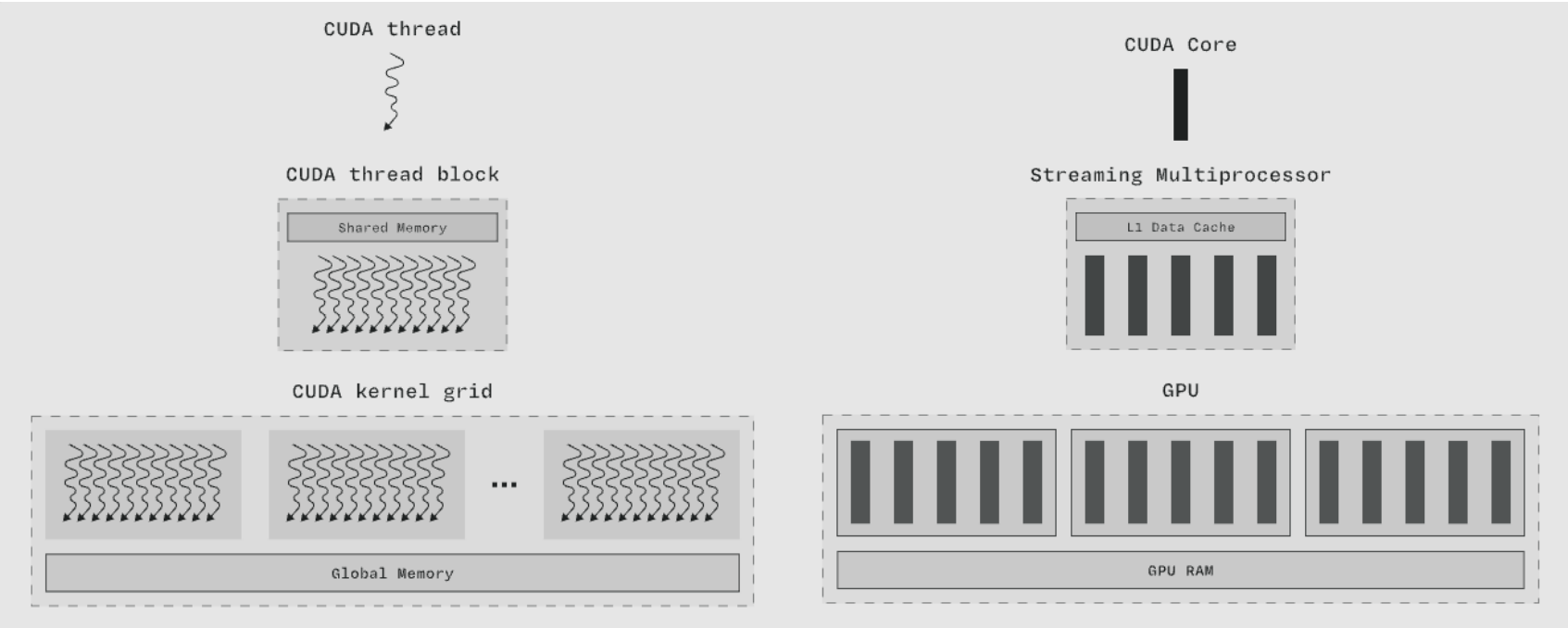


Image source: Modal, "CUDA Programming Model," *GPU Glossary*.

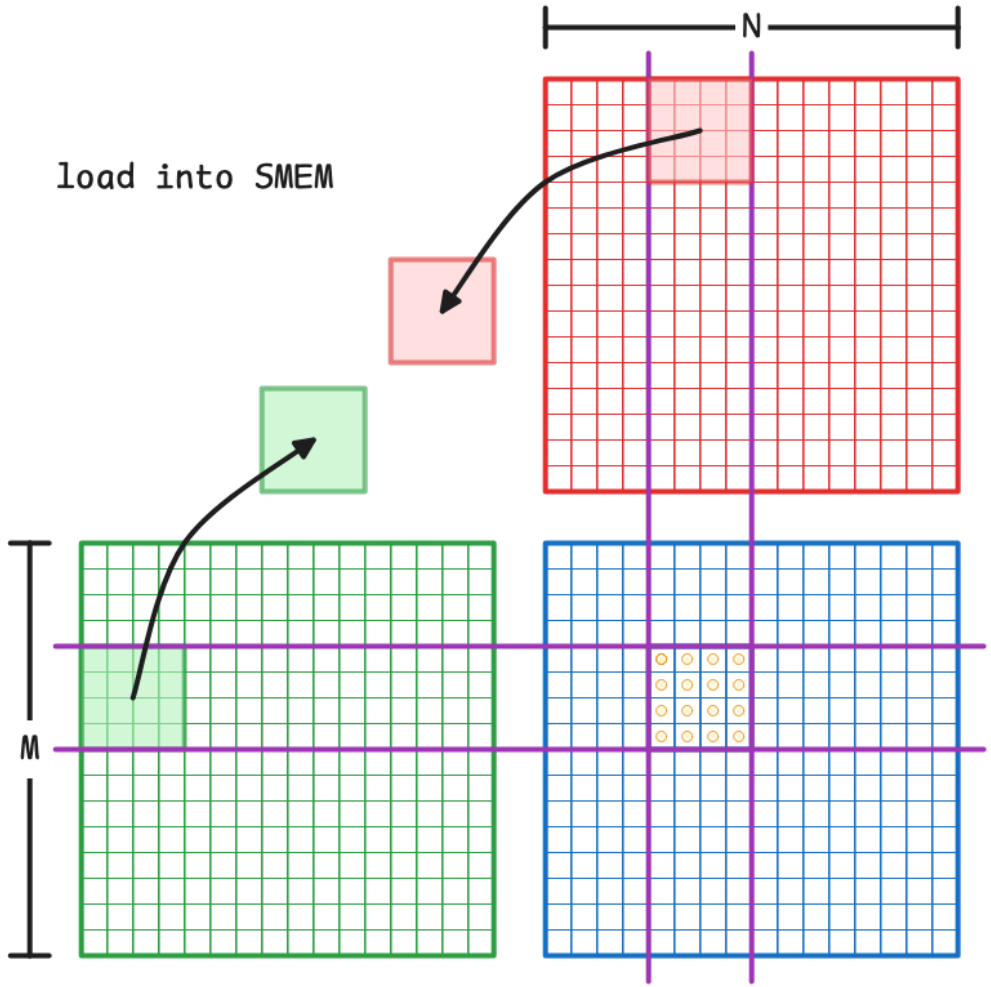
Tiling

Explicit caching of data shared between threads.

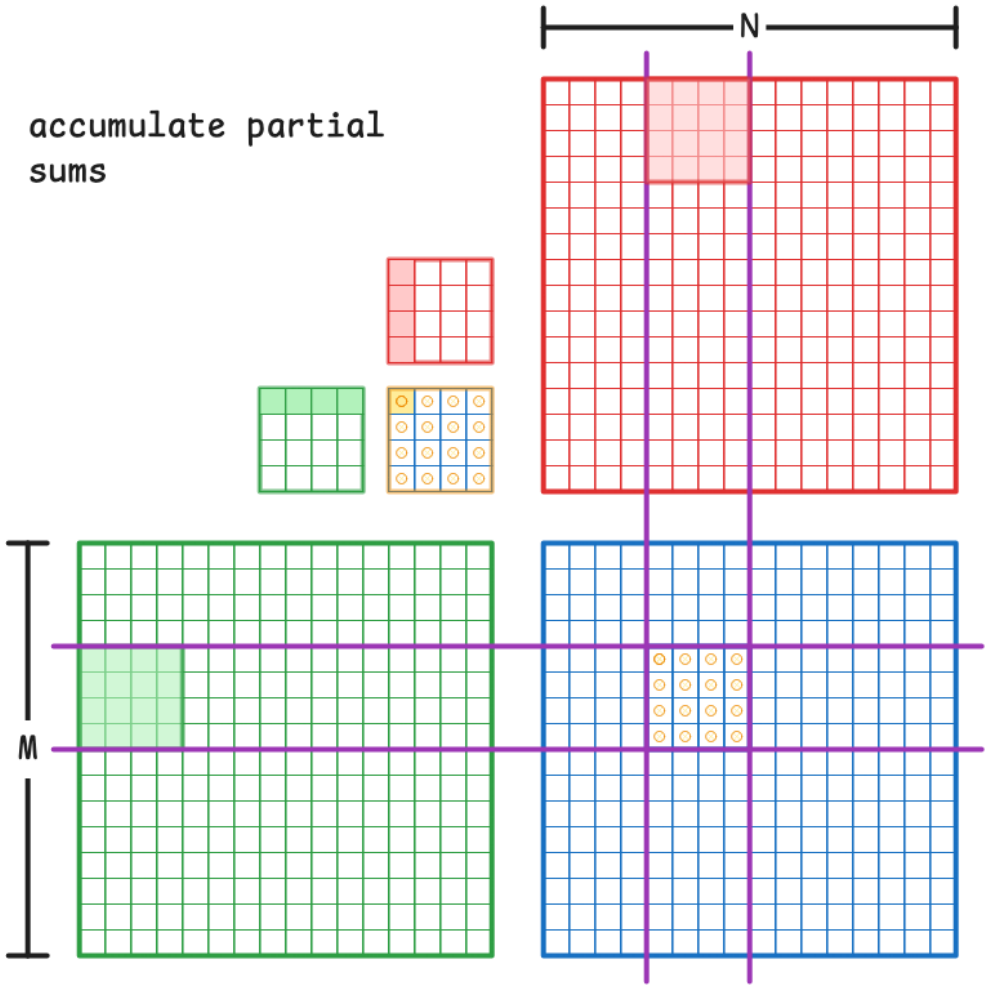
Main Idea

Before doing any computation, load a chunk of the necessary data from DRAM to SMEM as a block. Then, memory accesses for the computation now only go to SMEM.

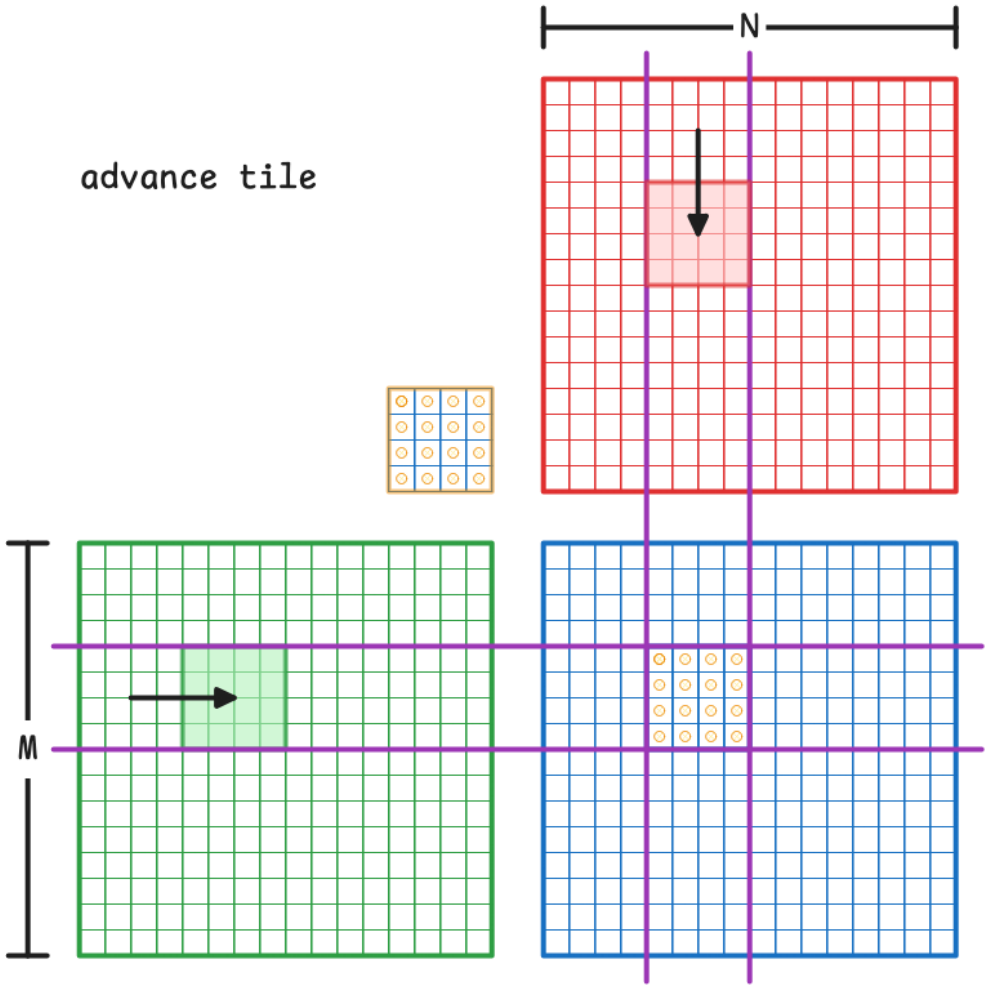
Tiling steps



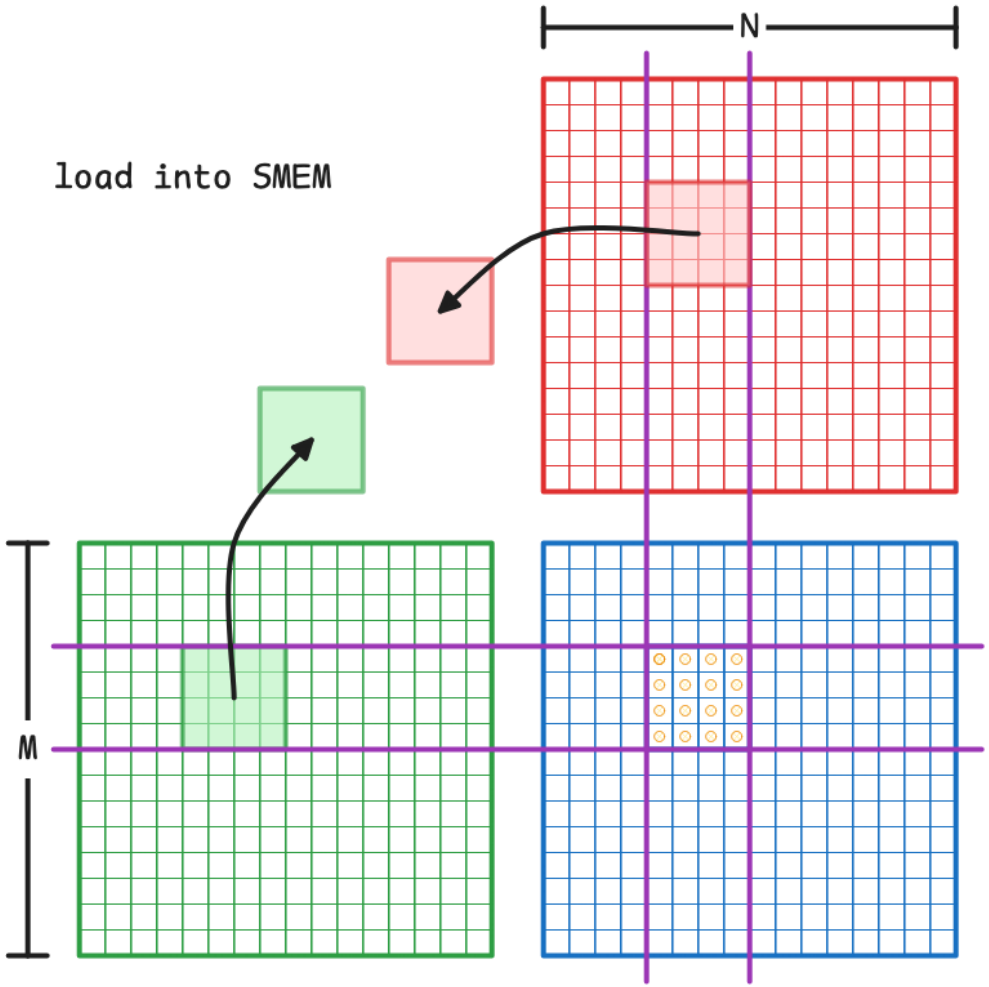
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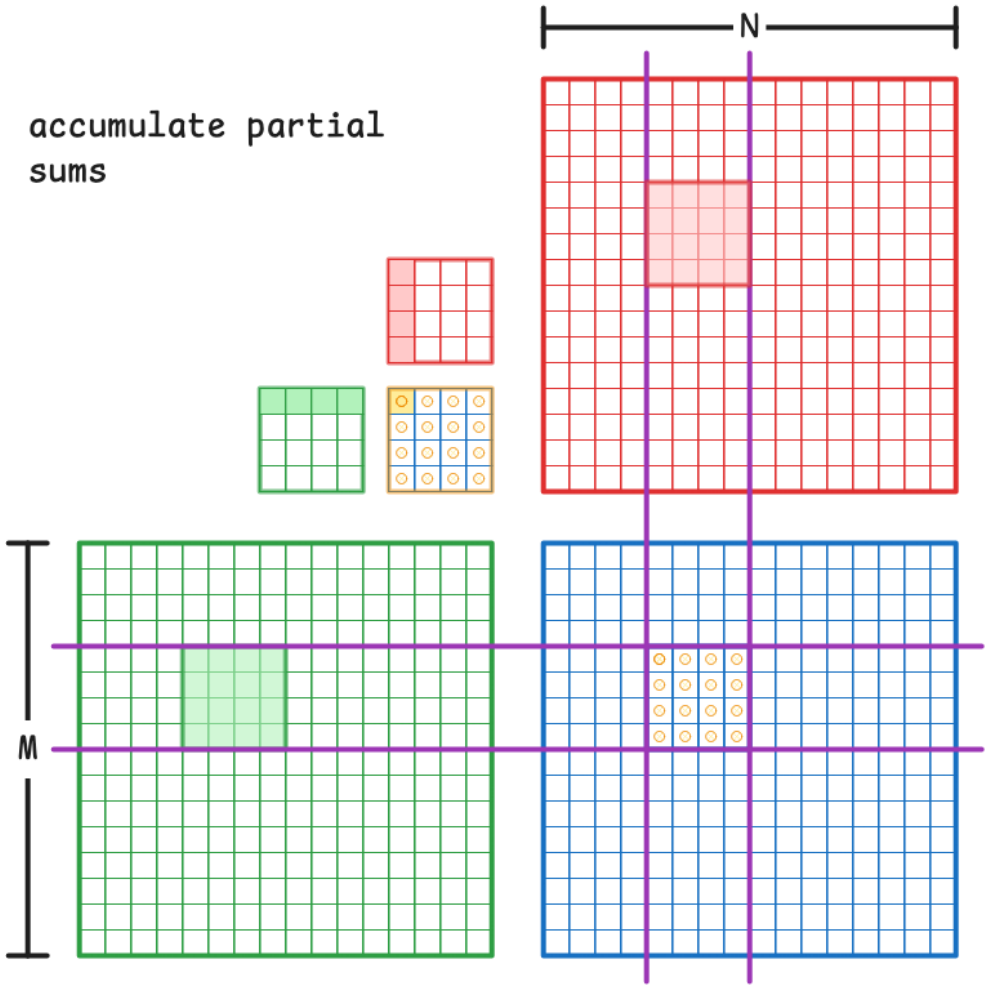
Tiling steps



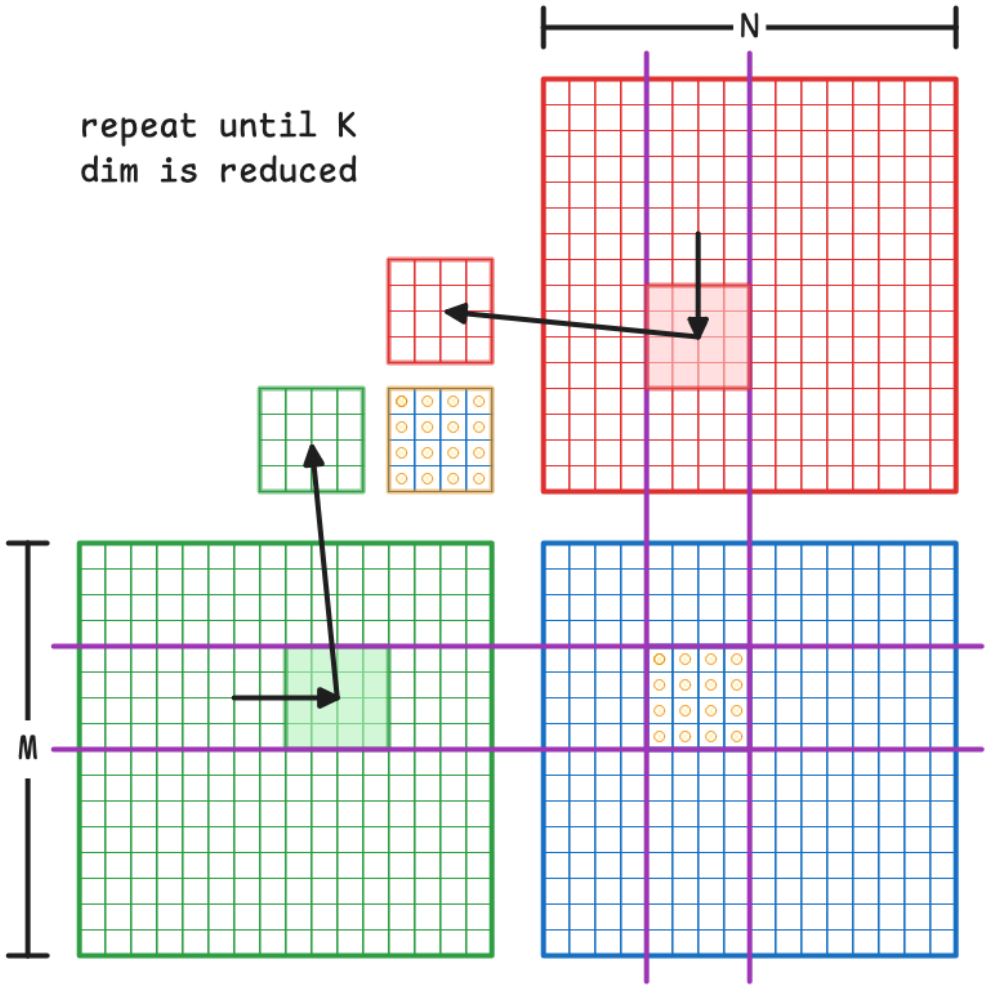
Tiling steps



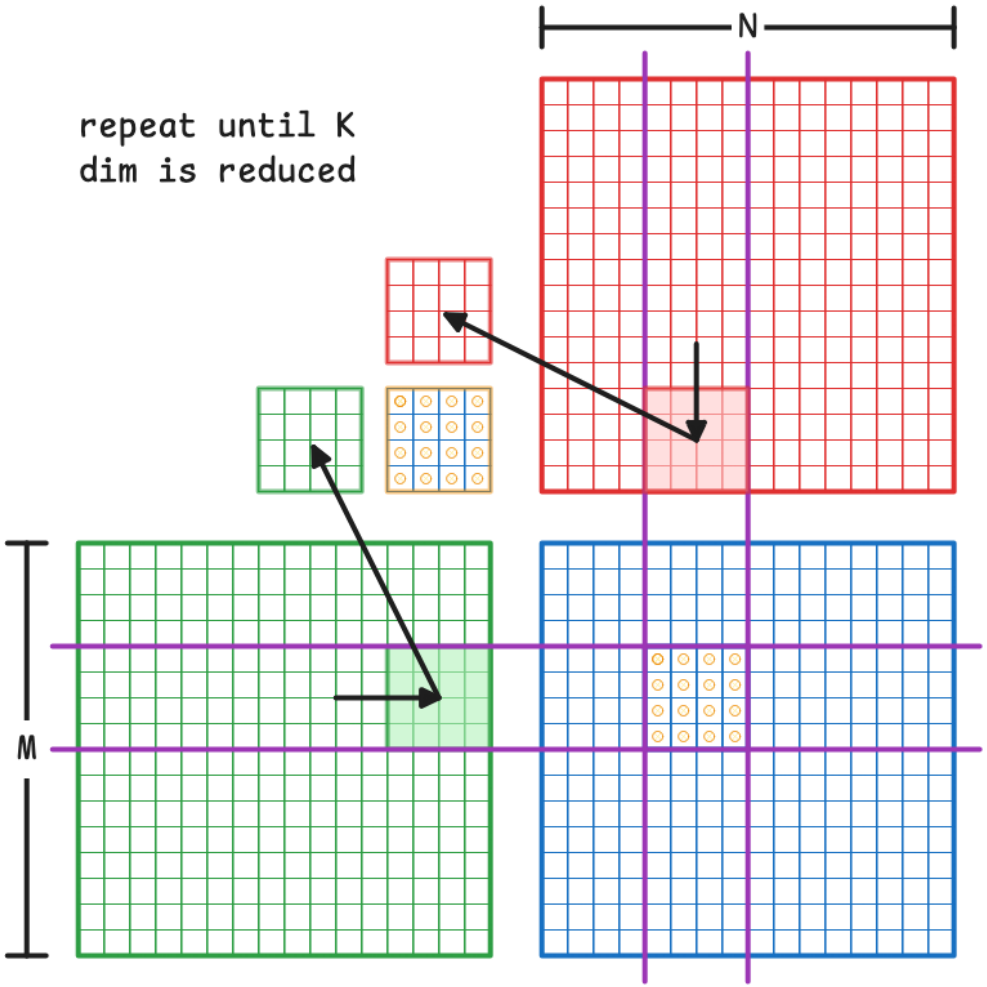
Tiling steps



Tiling steps



Tiling steps



Basic tiled SGEMM

```
__shared__ float As[BK][BK];
__shared__ float Bs[BK][BK];
float sum = 0.0;

for (int k = 0; k < K; k += BK) {
  const int A_tile_x = k + xt;
  if (y < M && A_tile_x < K) {
    As[yt][xt] = A[y * K + A_tile_x];
  } else {
    As[yt][xt] = 0.0;
  }

  const int B_tile_y = k + yt;
  if (B_tile_y < K && x < N) {
    Bs[yt][xt] = B[B_tile_y * N + x];
  } else {
    Bs[yt][xt] = 0.0;
  }

  __syncthreads();

  for (int i = 0; i < BK; i++) {
    sum += As[yt][i] * Bs[i][xt];
  }
  __syncthreads();
}
```

Outer loop:
Loop over tiles.

Cooperative loading to SMEM

Compute partial sum only with SMEM

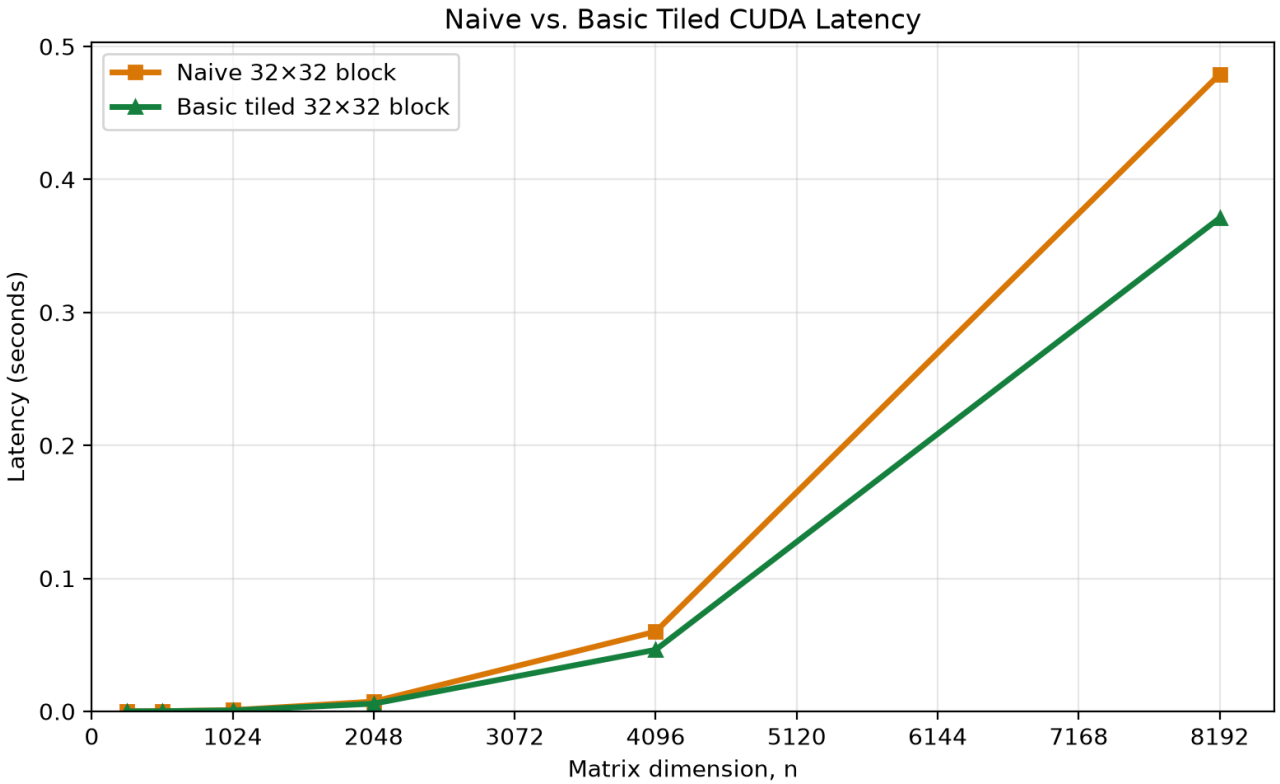
Source: `my_sgemm/src/sgemm.cu`.

Basic tiled SGEMM visualization

- Watch each block load small regions of A and B together.
- The loaded tiles are reused to update an entire tile of C.
- Synchronization separates loading, computing, and advancing to the next tiles.

Play tiled-kernel visualization

Tiling cuts latency by another order of magnitude



Can we still do better?

Throughput analysis

We report a final latency of 0.36s or 360ms for $n, m, k = 8192$. This is around $10\times$ the theoretical minimum of 31ms. This slowdown suggests we are still **memory bound**.

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This tiling setup with 32×32 blocks and $n, m, k = 8192$, we need a **grid** of $\frac{m}{32} \times \frac{n}{32} = 256 \times 256 = 65,536$ blocks.

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Thus, we have

$$\underbrace{65,536}_{\text{blocks}} \times \underbrace{256}_{\text{tiles/block}} \times \underbrace{2}_{A \text{ and } B} \times \underbrace{32 \times 32}_{\text{elements/tile}} \times \underbrace{4 \text{ bytes}}_{\text{bytes/element}} \approx 137.7 \text{ GB}.$$

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